

Daeho Lee

Education

- **Freie Universität Berlin** **GPA: 1.3/4 (1.0 best)**
Oct. 2024 – April.2026 , Berlin, Germany
M.S in Bioinformatics
- **Korea Advanced Institute of Science and Technology (KAIST)** **GPA: 3.98/4.3**
Sep. 2021 - Feb. 2024, Daejeon, S.Korea
M.S in the field of Bio and Brain Engineering
- **Sogang University** **GPA: 3.65/4.3**
Feb. 2016 - Aug. 2021 Seoul, S.Korea
B.S. in Life Science
Double Major in Artificial Intelligence (AI)

Research Experience

Computational and developmental biology

- Dr. Markus Mittnenzweig, Max Delbuck Center
- *Oct. 2025 - current, Berlin, German*
- *Research Student*
- Master's Thesis: Quantitative alignment of single-cell and spatial transcriptomics for cell-type mapping

Summary: Developed a two-stage alignment framework for mapping scRNA-seq profiles to 3D whole-embryo image-based spatial transcriptomics data. The method first performs global cell-to-spot matching using a quantitative similarity score, then applies signal-mixing-aware local refinement by removing boundary-associated mixing genes and optimizing the remaining expression structure with optimal transport.

Result: Compared with moscot, the method reduced unrealistic mixing near tissue boundaries, improved cluster separation, and increased the mean gene-wise Pearson correlation from 0.39 to 0.54. In leave-one-out validation, performance remained stable and consistently exceeded moscot, indicating that the framework did not rely on a small subset of genes.

Computational and developmental biology

- Dr. Markus Mittnenzweig, Max Delbuck Center
- *Jun. 2025 – Sep 2025, Berlin, German*
- *Research Student*
- Performance Comparison of Ambient RNA Removal Models for Single-Cell Data Integration in Mouse Embryo Development

Summary: Built a benchmarking pipeline for ambient RNA correction in mouse embryo scRNA-seq and systematically compared SoupX, DecontX, CellBender, and MCnoise across contamination removal, expression preservation, and sequencing-depth sensitivity.

Result: Identified DecontX as the best overall model, providing the most balanced removal of extraembryonic and erythroid ambient RNA while preserving housekeeping gene expression and showing the least bias with respect to sequencing depth.

Laboratory Data Science in the Life Sciences Class

- Prof. Dr. Katharina Jahn, Freie Universität Berlin, Department of Bioinformatics
- *Apr. 2025 - Sep. 2024, Berlin, German*
- *Research Student*
- Interpretable Glioma Classification via Cluster-Guided Multi-Omics Modelling

Summary: Conducted this project to address the limitation that histology-based labels such as LGG and GBM often fail to capture molecular heterogeneity and prognosis differences in glioma. Built a multi-omics analysis pipeline for TCGA LGG/GBM, including preprocessing, MOFA+ latent factor integration, consensus clustering, and cluster-specific classification.

Result: Identified three robust molecular glioma subgroups not captured by histology alone. Cluster-specific classifiers outperformed a global model, improving F1 from 0.892 to 0.964 on the internal test set and from 0.919 to 0.977 on the external test set, with clear survival differences across subgroups.

Single-Cell Medical Genomics Laboratory

- Prof. Jong-Eun Park, Graduate school of Medical Science and Engineering, KAIST
- *Dec. 2023 - Feb. 2024, Daejeon, S.Korea*
- *Research Student*
- Learn the Single Cell Tutorial and Research Single Cell-based Drug Development Tools

Summary: Conducted a structured review of bulk and single-cell drug response studies and designed a transfer-learning-based framework linking bulk expression, single-cell transcriptomes and drug-gene knowledge.

Result: Clarified why naïve bulk-based models fail to capture cellular heterogeneity and proposed a concrete workflow for bulk-single-cell integration to study resistance mechanisms.

Bio-Intelligence Class

- Prof. Dongsup Kim, Bio and Brain Engineering, KAIST
- *Mar. 2022 - Jul. 2022, Daejeon, S.Korea*
- *Research Student*
- Graph-based drug-target interaction prediction

Summary: Designed a drug-target interaction model on a DrugBank-based bipartite graph using ChemBERTa/ProtBERT embeddings, MultiSAGE and an MLP, with a focus on class imbalance.

Result: Showed that balanced negative sampling is essential to reach ~0.71 balanced accuracy and that some high-scoring predicted negatives correspond to biologically plausible interactions not annotated in the dataset.

Synergistic Bioinformatics Laboratory

- Prof. Gwan-Su Yi, Bio and Brain Engineering, KAIST
- *Sep. 2021 - Feb. 2024, Daejeon, S.Korea*
- Master's Thesis: Construction of a drug mechanism database based on text mining of bioassay literature information (Using PubChem Bioassay Database)

Summary: Developed a large-scale text-mining pipeline to extract targets, functions and regulatory effects from 1.6M+ PubChem BioAssay records and link them to compound activity tables.

Result: Produced a relational drug mechanism database that more than doubled usable target annotations and substantially extended compound–mechanism coverage beyond existing curated resources.

Spoken Language Laboratory

- Prof. Ji-hwan Kim, Artificial Intelligence, Sogang Univ.
- *Jan. 2021 - Aug. 2021, Seoul, S.Korea*
- *Research Student*
- Development of a language model (BERT, BART) to improve Korean speech recognizer performance

Summary: Implemented a unified KorQuAD fine-tuning pipeline to compare KoBERT and KoBART as stronger Korean language models relevant for downstream speech recognition.

Result: Found a small but consistent EM/F1 advantage for KoBART over KoBERT and showed how pre-trained Transformers address robustness limits of simple count-based language models.

CICAGo Lab

- Prof. Jon-Lark Kim, Artificial Intelligence, Sogang Univ.
- *Aug. 2020 - Dec. 2020, Seoul, S.Korea*
- *Research Student*
- Data-driven analysis of western blot quality

Summary: Curated about one year of western blot images and metadata and trained machine-learning models to classify signal quality into clear, absent and overexposed classes.

Result: Identified exposure time, protein loading and sample order as major drivers of blot quality, with feature-based models achieving ~0.75–0.80 validation accuracy.

CICAGo Lab

- Prof. Jon-Lark Kim, Artificial Intelligence, Sogang Univ.
- *Aug. 2020 - Dec. 2020, Seoul, S.Korea*
- *Research Student*
- CNN-based red blood cell detection and classification

Summary: Developed and tuned CNN models for red blood cell detection and classification in microscopy images and expanded limited training data via manual YOLO-format annotation.

Result: Manual expansion of high-quality bounding-box labels improved detection consistency and increased validation accuracy from ~0.75 to ~0.88 on a small, biased dataset.

Neurophotronics Lab

- Prof. Myung wan Choi, Life Science, Seoul National Univ.
- *Aug. 2020 - Jan. 2021, Seoul, S.Korea*
- *Research Student*
- Prediction of myelinated axons phenotype according to aging based on machine learning

Summary: Assisted two-photon imaging experiments on myelinated axons in mouse brain and constructed manually annotated datasets capturing axon location and morphological features across age groups. This work established an image-based data resource for analysing age-associated changes in axonal phenotypes.

Result: Generated a curated dataset suitable for initial machine-learning modelling and qualitative phenotype analysis. The project also showed that downstream model performance and interpretability were strongly influenced by imaging quality and annotation consistency.

Plant Molecular Biology and Biochemistry Laboratory

- Prof. Mi Chung Suh, Life Science, Sogang Univ.
- Jan. 2019 - Aug. 2020, Seoul, S.Korea
- Research Student
- Control of Metabolic Flux between Intracellular and Extracellular Lipids Biosynthesis

Summary: Re-analysed public seed RNA-seq data to prioritise KCS17 as a seed coat-enriched candidate and supported follow-up expression, localisation and lipid profiling experiments.

Result: Contributed to identifying KCS17 as an ER-localised enzyme required for very-long-chain fatty acid biosynthesis in flowers and seed coat waxes, with kcs17 mutants showing reduced VLCFAs and wax load.

Project Experiences

- Development and validation of AI-based drug repositioning funded by the MOHW of the Ministry of Science and ICT (MSIT), South Korea, Sep. 2021 - Jan. 2022
- Identification of drug candidate based on AI text mining funded by KAIST, South Korea Sep. 2022 - Feb. 2023

Publication

- Jun young Park, **Daeho Lee**, Bang Yoon, Na yeon Lee, & Ji-hwan Kim. (2021). [Performance Analysis of KoBART Model Using KorQuAD Dataset and KoBERT Tokenizer.](#) *Proceedings of the Korean Information Science Society Conference*, 1747-1749.

Honors and Awards

- 2021 Excellence Award in the undergraduate category, Korea Information Computer Congress (KCC)

Teaching Experience

- TA in Department of Bio and Brain Engineering, KAIST, Daejeon, Korea, Mar. 2023 - Aug. 2023
- Course: Bioengineering Lab 1: Microarray data analysis using Decision Tree
- TA in Department of Bio and Brain Engineering, KAIST, Daejeon, Korea, Sep. 2022 - Dec. 2022
- Course: Biomedical Statistics & Statistical Learning
- TA in Department of Bio and Brain Engineering, KAIST, Daejeon, Korea, Mar. 2022 - Aug. 2022

- Course: Bioengineering Lab 1: Microarray data analysis using Decision Tree

Skills

- Proficient in Python, R and SQL programming
- Deep experience of statistical analysis methods and machine learning techniques
- Familiarity with Unix/Linux systems
- Experience in C, C++ programming